Impact Modeling of Judicial system on Drivers Behaviour via Evolution Game Theory

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***Abstract*— The article considers the issue of modeling a complex socio-psychological system in which there are two subsystems: an accident modeling subsystem that describes the dynamics of accidents and driver behavior expressed in the possibility of a driver violating the speed limit, driving drunk or without a seat belt, as well as a deterministic judicial subsystem that takes into account the circumstances of the incident and the vector of regulatory policy. On the basis of which a decision is made on the imposition of a fine, deprivation of rights for a fixed period and sentencing to prison, based on data from judicial practice (court decisions) under Law 12.8 of the Administrative Code of the Russian Federation and Law 264.1 of the Criminal Code of the Russian Federation. This paper also presents a series of numerical experiments aimed at assessing the impact of changes to the lower and upper limits of license revocation under Article 12.8 of the Administrative Code, as well as a series of stress tests to evaluate the impact of the likelihood of driving under the influence and the cost of traffic accidents on the overall dynamics of the incident subsystem.**

Keywords — government regulation, traffic policy, EGT, judicial system, driver's behavior, fastened driving, speeding, drunk driving

I. Introduction

The problem of modeling systems of criminal incidents and legal proceedings is a common task within the Legaltech sector. The most widespread studies focus on predicting incidents [1], crime spots forecasting [2], criminal recidivism prediction [3] etc. On the other hand, the issue of assessing and modeling various forms of regulation in the legal sphere remains understudied. Based on the previously presented concept [4], the article proposes expanding the current functionality of the methodology by adding a new class of incidents (driving under the influence) and incorporating a legal proceedings system that allows building a model based on the circumstances of incidents and the overall judicial practice across the country. The current model, based on a combination of evolutionary game theory methods, co-evolutionary algorithms, and methods for predicting court decisions based on the **Histogram-based Gradient Boosting Classification Tree** and **Histogram-based Gradient Boosting Regression Tree** methods, allows using the resulting system to assess the dynamics of incidents that depend on national-specific parameters and parameters of government regulatory policy.

The article presents research into the dynamics of road accidents related to speeding, driving under the influence, and not wearing a seatbelt, as the most illustrative examples within the framework of a model that includes strict and mild forms of government policy in the field of road safety. The current methodology makes it possible to track how various measures and the degree of regulatory strictness affect the trends and dynamics of key system performance indicators.

II. Methods

Within the section under consideration, a methodology for traffic policy modeling using evolutionary game theory is presented. Within the model, two components can be distinguished: the road environment system and the judicial system. Before examining each of them separately, we will define the key limitations of the model.

A. Model limitations

• It is believed that when driving under the influence, the driver does not commit violent offenses (this allows us to consider only Articles 12.8 of the Administrative Code and 264.1 of the Criminal Code within the judicial system);

• The cost of traffic accidents in a country with strict and lax regulation is the same;

• The amount of fines for exceeding the speed limit is the same and does not depend on how much the limit was exceeded (there is no progressive scale);

• The likelihood of a traffic accident and a fatal outcome increases by the same multiple if the driver is intoxicated;

• A driver who has had their license revoked continues to pay taxes.

• A driver who has had their license revoked will not be allowed to drive until it is restored;

• The duration of one era is 180 days;

• A drunk driver retains the chosen driving strategy class

B. The methodology of the "road environment" system

The key concept of the road environment system is the set of agents of the driver type ("D") and the state type ("G"). Based on the results of the available literature, it was established that the indicator of travel success for agents of type D is travel time [5], and for agents of type G, it is the average expenditure [6]. During initialization there are N_{driver} type D and type G agents.

During initialization, each agent D is assigned a driving strategy (exceeding or adhering to the speed limit) with equal probability, as well as a seatbelt status indicator *belted,* speed

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of movement *speed* in km/h from the interval speed^{o}_{min} и speed^{o}_{max}, if the driver complies with the speed limit, and from the interval speed^{s}_{min} and speed^{s}_{max}, if the driver exceeds the speed limit, and there's a risk of drunk driving *dr* from the interval between dr_{min} and dr_{max}. Additionally, the expected travel time is initialized for each driver d_{exp}, distributed according to the normal distribution with parameters mean μ_{exp} and standard deviation σ^2_{exp}.

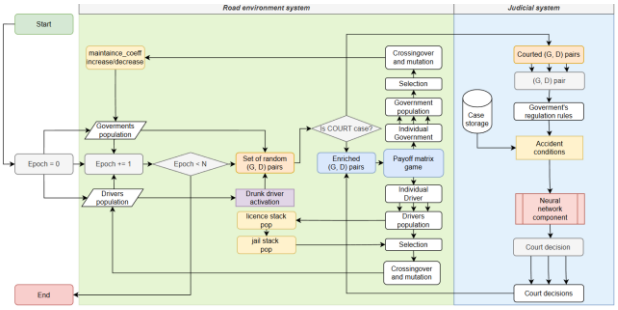


Fig. 1. Scheme for modeling an experiment in a system for tracking the dynamics of driver behavior and government policies.

During initialization, each agent G is assigned with equal probability to either a strict ("Mild") or a mild state policy strategy, along with the following parameters: the initial condition factor of the road θ_{ma} with a fixed value θ^{s}_{ma} (θ^{m}_{ma}); the random fine amount for driving without a seatbelt between feeB^{s}_{min} ($feeB^{m}_{min}$) and $feeB^{s}_{max}$ ($feeB^{m}_{max}$) для «Строгий» ("Mild") policies in rubles; a random fine for exceeding the speed limit feeS^{s}_{min} ($feeS^{m}_{min}$) and $feeS^{s}_{max}$ ($feeS^{m}_{max}$) for "Strict" ("Mild") in rubles; fixed lower and upper limits of fines $feeD^{s}_{min}$ ($feeD^{m}_{min}$) and $feeD^{s}_{max}$ ($feeD^{m}_{max}$) in rubles; the chance of getting into a traffic accident $crash_s$ ($crash_m$); chance of the fatal outcome $fatal_s$ ($fatal_m$); The parameter representing the average cost per drivertr_s (tr_m) for the implementation of the chosen regulatory policy; the chance of being stopped by the police $check_s$ ($check_m$); The duration of the vehicle inspection d_s (d_m); The average cost of keeping a driver in prison during that era c^{jail}_{s} (c^{jail}_{m}). In addition, the general parameters of the legal framework under Articles 12.8 and 264.1, in terms of the lower and upper limits of the prison term in years, the minimum and maximum limits of fines, and the minimum and maximum periods of license revocation..

Also, before launching the experiment, the parameters of national specifics are set: the length of the *len* road in km; reducing the coefficient of trip duration during drunk driving. θ_{dr} and if the speed limit is exceeded θ_d; The coefficient that increases the likelihood of a traffic accident and fatal outcome in cases of drunk driving f_{dr}; the increasing factor for speed limit violations leading to traffic accidentsθ_{cr} and for the fatal outcome θ_{fc}; the perception coefficient of conditional punishmentsθ_{con}; perception coefficient of prison sentences θ_j; the perception coefficient of driving license revocation θ_l; the vehicle inspection perception coefficient θ_{ch}; the perception coefficient of government spending θ_{sp}; The coefficient of perception of the imposed fine θ_{fee}; The reduction factor when wearing a seatbelt f_{min}.

After initializing the model, the system proceeds to launch a series of N_{epoch} experiments. At the beginning of each epoch, with an individual chance *dr* the type D agent can drive while being intoxicated. Next, from the set of active drivers at the beginning of the epoch N_{active} (for drivers who have not had their licenses revoked and are not in prison) the corresponding number of "D"–"G" pairs is generated. After that, the rewards are calculated according to Table 1. If the driver was drunk (the "drunk" flag = 1) and either got into a traffic accident or was stopped by the police, the system sends the bar to the judicial system, from which the court's decision on the duration of detention in prison for driving under the influence is received d^{cou}_{j} in hrs; the period of license revocation for drunk driving d^{cou}_{l} in hrs; the sign of a suspended sentence COND; the amount of the fine imposed for drunk driving $feeD$ in RUB; the amount of court's compensation $feeLeg$ in RUB. The number of Type D agents in prison N^{epoch}_{jailed} and type "D" agents who continue to bear the tax burdenN^{epoch}_{driver} (it includes drivers who are not serving their sentences and drivers who have had their licenses revoked but are not in prison) is recalculated at the beginning of each era.

TABLE I. PAYOFF MATRIX

Government's Regulation strategy	Driver's speed table	
	Obeying speed limit	***Speeding***
Strict regulation	GSO, DSO	GSS, DSS
Mild regulation	GMO, DMO	GMS, DMS

The calculation of remuneration in hours for drivers is based on the following formulas:

$$DSO = d_{exp} - \frac{\text{len}}{speed \cdot \theta_{ma} \cdot (1 + drunk \cdot \theta_{dr})} - \\ - \text{Bern}(\text{check}_s) \cdot d_s \cdot \theta_{ch} - \\ - \text{Bern}(p^{so}_{fatal}) \cdot 10^7 - \\ - \text{Bern}(p^{so}_{crash}) \cdot (N(\mu_s, \sigma^2_s) + d_s \cdot \theta_{ch}) - \\ - \text{COURT} \cdot (d^{cou}_{j} \cdot \theta_j + d^{cou}_{l} \cdot \theta_l) \cdot (1 - \text{COND} \cdot \theta_{con}); \quad (1)$$

$$DMO = d_{exp} - \frac{\text{len}}{speed \cdot \theta_{ma} \cdot (1 + drunk \cdot \theta_{dr})} - \\ - \text{Bern}(\text{check}_m) \cdot d_m \cdot \theta_{ch} - \\ - \text{Bern}(p^{mo}_{fatal}) \cdot 10^7 - \\ - \text{Bern}(p^{mo}_{crash}) \cdot (N(\mu_m, \sigma^2_m) + d_m \cdot \theta_{ch}) - \\ - \text{COURT} \cdot (d^{cou}_{j} \cdot \theta_j + d^{cou}_{l} \cdot \theta_l) \cdot (1 - \text{COND} \cdot \theta_{con}); \quad (2)$$

$$DSS = d_{exp} - \frac{\text{len}}{speed \cdot \theta_{ma} \cdot (1 + drunk \cdot \theta_{dr})} \cdot \theta_d - \\ - \text{Bern}(\text{check}_s) \cdot d_s \cdot \theta_{ch} - \\ - \text{Bern}(p^{ss}_{fatal}) \cdot 10^7 - \\ - \text{Bern}(p^{ss}_{crash}) \cdot (N(\mu_s, \sigma^2_s) + d_s \cdot \theta_{ch}) - \\ - \text{COURT} \cdot (d^{cou}_{j} \cdot \theta_j + d^{cou}_{l} \cdot \theta_l) \cdot (1 - \text{COND} \cdot \theta_{con}); \quad (3)$$

$$DMS = d_{exp} - \frac{\text{len}}{speed \cdot \theta_{ma} \cdot (1 + drunk \cdot \theta_{dr})} \cdot \theta_d - \\ - \text{Bern}(\text{check}_m) \cdot d_m \cdot \theta_{ch} - \\ - \text{Bern}(p^{ms}_{fatal}) \cdot 10^7 - \\ - \text{Bern}(p^{ms}_{crash}) \cdot (N(\mu_m, \sigma^2_m) + d_m \cdot \theta_{ch}) - \\ - \text{COURT} \cdot (d^{cou}_{j} \cdot \theta_j + d^{cou}_{l} \cdot \theta_l) \cdot (1 - \text{COND} \cdot \theta_{con}) \quad (4)$$

The calculation of remuneration for type G agents in rubles is based on the following formulas:

$$GSO = (-\mathrm{c} \cdot \mathrm{Bern}(p_{crash}^{so}) - tr_s) \cdot \theta_{sp} + \\ +(\mathrm{belted} \cdot \mathrm{feeB}_s) \cdot \theta_{fee} \cdot \mathrm{Bern}(\mathrm{check}_s) + \\ + \mathrm{COURT} \cdot (\mathrm{feeD} \cdot \theta_{fee} - \mathrm{feeLeg} \cdot \theta_{sp}) - \\ - \frac{N_{jailed}^{epoch}}{N_{driver}^{epoch}} \cdot c_s^{jail} \cdot \theta_{sp}; \quad (5)$$

$$GSS = (-\mathrm{c} \cdot \mathrm{Bern}(p_{crash}^{ss}) - tr_s) \cdot \theta_{sp} + \\ + (\mathrm{belted} \cdot \mathrm{feeB}_s + \mathrm{feeS}_s) \cdot \theta_{fee} \cdot \mathrm{Bern}(\mathrm{check}_s) + \\ + \mathrm{COURT} \cdot (\mathrm{feeD} \cdot \theta_{fee} - \mathrm{feeLeg} \cdot \theta_{sp}) - \\ - \frac{N_{jailed}^{epoch}}{N_{driver}^{epoch}} \cdot c_s^{jail} \cdot \theta_{sp}; \quad (6)$$

$$GMO = (-\mathrm{c} \cdot \mathrm{Bern}(p_{crash}^{mo}) - tr_m) \cdot \theta_{sp} + \\ + (\mathrm{belted} \cdot \mathrm{feeB}_m) \cdot \theta_{\mathrm{fee}} \cdot \mathrm{Bern}(\mathrm{check}_m) + \\ + \mathrm{COURT} \cdot (\mathrm{feeD} \cdot \theta_{fee} - \mathrm{feeLeg} \cdot \theta_{sp}) - \\ - \frac{N_{jailed}^{epoch}}{N_{driver}^{epoch}} \cdot c_m^{jail} \cdot \theta_{sp}; \quad (7)$$

$$GMS = (-\mathrm{c} \cdot \mathrm{Bern}(p_{crash}^{ms}) - tr_m) \cdot \theta_{sp} + \\ + (\mathrm{belted} \cdot \mathrm{feeB}_m + \mathrm{feeS}_m) \cdot \theta_{\mathrm{fee}} \cdot \mathrm{Bern}(\mathrm{check}_m) + \\ + \mathrm{COURT} \cdot (\mathrm{feeD} \cdot \theta_{fee} - \mathrm{feeLeg} \cdot \theta_{sp}) - \\ - \frac{N_{jailed}^{epoch}}{N_{driver}^{epoch}} \cdot c_m^{jail} \cdot \theta_{sp}; \quad (8)$$

The probability of a fatal outcome occurring under various combinations of strategies for pairs of agents "D" and "G" is calculated using the following formulas:

$$p_{fatal}^{so} = \mathrm{fatal}_s \cdot (1 + drunk \cdot \mathrm{f}_{dr}) \cdot \\ \cdot (1 - belted \cdot (1 - \mathrm{f}_{min})); \quad (9)$$

$$p_{fatal}^{mo} = \mathrm{fatal}_m \cdot (1 + drunk \cdot \mathrm{f}_{dr}) \cdot \\ \cdot (1 - belted \cdot (1 - \mathrm{f}_{min})); \quad (10)$$

$$p_{fatal}^{ms} = \mathrm{fatal}_s \cdot \theta_{fc} \cdot (1 + drunk \cdot \mathrm{f}_{dr}) \cdot \\ \cdot (1 - belted \cdot (1 - \mathrm{f}_{min})); \quad (11)$$

$$p_{fatal}^{ms} = \mathrm{fatal}_m \cdot \theta_{fc} \cdot (1 + drunk \cdot \mathrm{f}_{dr}) \cdot \\ \cdot (1 - belted \cdot (1 - \mathrm{f}_{min})). \quad (12)$$

The probability of a crash outcome occurring under various combinations of strategies for pairs of agents "D" and "G" is calculated using the formulas:

$$p_{crash}^{so} = crash_s \cdot (1 + drunk \cdot \mathrm{f}_{dr}) \quad (13)$$

$$p_{crash}^{mo} = crash_m \cdot (1 + drunk \cdot \mathrm{f}_{dr}) \quad (14)$$

$$p_{crash}^{ss} = crash_s \cdot \theta_{cr} \cdot (1 + drunk \cdot \mathrm{f}_{dr}) \quad (15)$$

$$p_{crash}^{ms} = crash_m \cdot \theta_{cr} \cdot (1 + drunk \cdot \mathrm{f}_{dr}) \quad (16)$$

After receiving the rewards, the system ranks N_{active} and ranks agents of type G and D within their own sets.

During the selection stage, the system selects half of the best agents of types "D" and "G" (hereinafter referred to as the selected agents). Next, the number of active drivers N_{new}: The number of convicted drivers is subtracted from the size of the previous population (drivers with suspended sentences are not included), and drivers whose driving license revocation and prison sentence have expired are added. Based on the previously selected agents, the system forms a new population of the specified sizeN_{new} through multiple duplication of the selected agents (first through duplication of the selected sample, and then through random selection of elements from the selected sample) or, if $N_{new} < N_{active}$, by reducing the number of selected agents in the original population to half the number N_{new} in accordance with the ranking list, after which the resulting set of agents is duplicated. In addition, the "Prisons" stack and the "License Revocation" stack are updated: new agents are added to them with a specified term of imprisonment; the term of imprisonment for agents previously present in the system is reduced by 1 epoch. If the value becomes 0, then the agent is removed from the stack.

During the crossing-over stage, an equal probability exchange of genes occurs within the formed populations "D" and "G". In population "D", randomly selected pairs "D"–"D" exchange the belted trait, the chance of drunk driving dr, or proportionally exchange speeds. In population "G", randomly selected pairs "G"–"G" exchange the infrastructure condition traits θ_{ma} or proportionally exchange the amount of fines for speeding *feeS* or for driving without a seatbelt *feeB*.

Next, at the mutation stage, for the resulting set of agents "D" and "G", a mutation chance m occurs, respectively. With a given mutation threshold, agent "D" may change its strategy (in which case all driver parameters will be proportionally replaced), or its speed, or the value of the drunk driving chance dr, or the value of the seatbelt driving indicator *belted* to the opposite. With a given mutation threshold, agent "G" may change its strategy (in which case all penalty parameters are proportionally replaced), or the amount of fines for speeding *feeS*, or for driving without a seatbelt *feeB*, or the infrastructure indicator θ_{ma} set in the range of [0.4; 1].

At the end of an epoch, for all agents "G," the parameter θ_{ma} increases by Δ_{ma}, if the agent has a strict policy, or decreases by Δ_{ma}, if the agent has a mild policy. After that a new epoch begins. When the epoch counter get to N_{epochs} system will stop the experiment.

C. Методология the judicial system

Will the judicial system be built for those "G"–"D" pairs in which agent "D" has the *drunk*=1 flag and the agent himself has either been involved in a traffic accident or was stopped by the police. Then the system receives a vector of the current regulation from agent "G" and a random vector of the traffic accident circumstances, taken from a dataset compiled from the rulings of district and magistrate courts under Articles 12.8 and 264.1, based on the source [7]. Regulatory policy can include the left and right boundaries of the amount of the fine imposed for each article separately in rubles, the left and right boundaries of the license revocation period in years, as well as the left and right boundaries of the prison sentence period in years. The set of circumstance vectors consists of incidents that were not included in either the test or the training dataset. The total dataset of decisions under Article 12.8 of the Administrative Code occurs 70,615 times, and under Article 264.1 of the Criminal Code of the Russian Federation — 21,207 times.

The system's output values are (targets):

- the number of epochs of imprisonment for driving under the influence;
- the number of epochs of license revocation;
- whether the punishment is conditional COND;
- the amount of the imposed fine in rubles $feeD$;
- the amount of costs that the state will pay in rubles $feeLeg$.

For each target, a separate model is trained using the same vector of circumstances. For example, one model predicts the amount of a fine, another predicts the duration of driving license suspension, and a third predicts the conditional nature of the punishment. The values of the other targets are not added to the input. For a numerical target, the Histogram-based Gradient Boosting Regression Tree method is used; for a binary or categorical target, the Histogram-based Gradient Boosting Classification Tree method is used. The method is convenient as a tabular baseline: it models nonlinear dependencies and allows each target to be analyzed separately. However, the models do not coordinate their appearances with each other.

To assess the success of target prediction in class/binary classification, the F1-macro metric was used, and to assess the success of target regression prediction:

$$score_{reg} = \max(0, \quad screg); \tag{17}$$

$$screg = 1 - \frac{MAE}{mean\,(|y_{true}|) + std(y_{true}) + 1} \tag{18}$$

The final metric is the average value across all F1-macro and $score_{reg}$ target. As a result of a series of experiments, it was found that the prediction accuracy of the selected approach for the key metric was 0.987. Oversampling of the original sample did not lead to an increase in the model's accuracy.

III. Numerical simulation

Based on the model described earlier, we will conduct a series of numerical experiments, but first we will initialize the model.

A. Model initialization

As part of our study of the dynamics of road safety, we classify Russia and Norway as "Mild" and "Strict" countries, respectively, based on the safety rating of European countries [8,9]. The conversion from NOK to RUB was carried out based on the assumption that 1 NOK is equivalent to 8.66 RUB.

As part of the experiment, a 100-km long route outside the city with a speed limit of 100 km/h is being considered. A driver who complies with the speed limit can maintain a speed between 70 and 100 km/h at initialization. A driver who exceeds the speed limit can accelerate from 110 to 150 km/h at initialization. The probability that a driver may drive while intoxicated is uniformly distributed between 0 and 0.1 upon initialization. The cost of maintaining a driver for an entire period (180 days) amounted to 361,532 rubles in Russia and 5,839,841 rubles in Norway, based on the study data [10-12]. The probability of a fatal outcome on the road in Russia was 0.028%, based on the number of fatal traffic accidents in 2025 [13] and the number of registered passenger cars [14]. The probability of a fatal traffic accident in Norway, according to the report, [15] was 0.0016%. The probability of a traffic accident in Russia was 0.259%, based on the ratio of the number of accidents [13] to the number of registered vehicles [14]. For Norway, the probability of 0.01478% was calculated proportionally. For all policies, the average waiting time for the police was 0.27 hours, with a standard deviation of 0.25 hours, according to the study [16]. The probability of a vehicle being inspected was set at 60% in Norway and 40% in Russia, in accordance with the safety rating [8-9]. The cost of implementing the safe driving policy per driver in Russia amounted to 5,740 rubles, calculated based on the ratio of the cost of maintaining road infrastructure in 2025 [17] and Police force budgeting in 2024 [18] to the population in the same year [19]. In Norway, the cost of implementing the policy per driver was 177,722 rubles, taken as the sum of Norway's annual expenditures on Police service and Transport service in 2023 [20] divided by the population [11]. The fine for exceeding the speed limit in the described case is based on Article 12.9 of the Administrative Offenses Code [21] and ranges from 750 to 2,250 rubles. For Norway, the values were set at 9,000 to 126,900 rubles [22]. The lower limit of the fine for driving without a seat belt in Russia is 1,500 rubles [21], while in Norway it is 11,700 rubles [22]. The upper limit of the fine interval was taken as twice the value of the lower limit. The left and right limits for fines under Article 264.1 of the Criminal Code of the Russian Federation were set at 200,000 and 300,000 rubles in Russia [23], corresponding to 1.5 and 2 monthly salaries in Norway for strict state policy [24]. The left and right limits for the fine under Article 12.8 in Russia were set at 45,000 rubles [21] and a proportionally corresponding amount of 181,210 rubles for Norway. The remaining regulatory parameters were based on the Russian legal framework: under Article 12.8, the minimum and maximum terms of deprivation of rights are 1.5 and 2 years; under Article 264.1, the minimum and maximum terms of deprivation of rights are 0 and 3 years; under Article 264.1, the minimum and maximum terms of imprisonment are 2 years. The infrastructure condition indicators for the "Mild" policy were 0.6, and for the "Strict" policy — 0.8 [8-9]. The mutation coefficient is set to m=20%. The cost of traffic accidents in Russia is calculated as the ratio between the total losses for 2024, taken from the Decree of the President of the Russian Federation "On the Approval of the Strategy for Improving Road Safety in the Russian Federation for the Period up to 2030 and for the Long-Term Perspective up to 2036" [25] and the number of accidents for the same year [13]. The parameter for the average expected travel time is 1 hour, with a root-mean-square deviation of 0.1 hour. The minimization of fatal outcomes from wearing a seatbelt is determined to be 60%.

The parameters of the national specifications were selected to approximate three target indicators: the percentage of people driving with a seatbelt is 83% [26]; the percentage of people driving exceeding the speed limit is 38.45% [26]; the percentage of people driving under the influence is 0.3794% (calculated as the ratio of the number of traffic violations for drunk driving [27] to the total number of passenger car drivers [14]).

Within the framework of the experiment, the population $N_{drivers} = 1000$. Key indicators are calculated as the average over 10 runs for 300 epochs. Parameter $\Delta_{ma} = 0.1$.

With the established parameters, it was achieved that, on average, 79.8% of drivers in the model wear a seatbelt (with

an average deviation of 2.9%), 39.8% exceed the speed limit (with an average deviation of 1.07%), and 0.49% drive while intoxicated (with an average deviation of 0.02%).

B. Experiment

Based on the calibrated model, a series of tests was conducted. To begin with, the results of the experiment were recorded without changes to the regulation parameters and, using the previously specified parameters reflecting national specifics, a general graph of the dynamics of the system's key indicators over 300 epochs was created (fig. 2). These indicators included: the percentage of drivers wearing seat belts per epoch; the percentage of drivers violating the speed limit per epoch; the average number of prisoners per epoch; the average number of active drivers per epoch; the percentage of G-type agents with strict state policy; the percentage of drivers without a license; the percentage of drunk drivers per epoch and the average speed per epoch.

Next, an experiment was conducted to assess how a change in the range of terms of deprivation of rights under Article 12.8 in years entails a change in the dynamics of key indicators (fig. 3). At the same time, we note that the value of the left and right boundaries in the experiment was changed and replaced by both strict and soft regulatory policies.

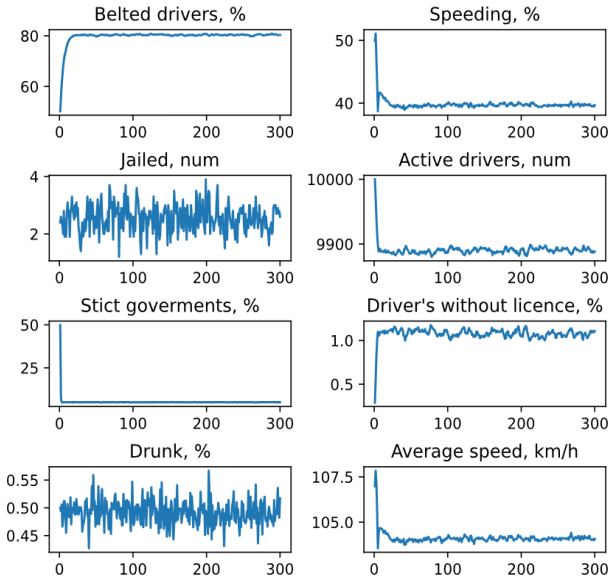


Fig. 2. Graphs showing key indicators under basic model settings

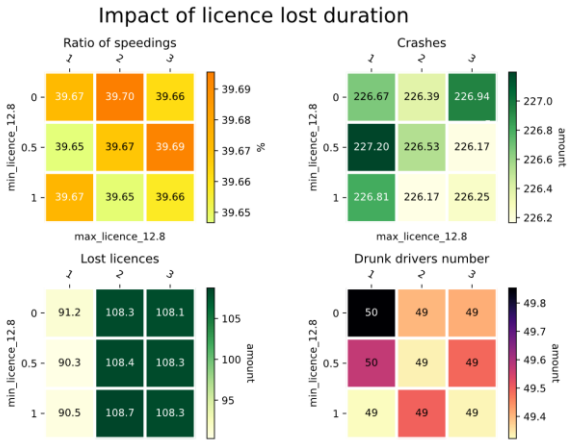


Fig. 3. Heat map showing changes in key indicators when the terms of license revocation are altered

Наконец было проведено два стресс-теста. В первом вероятность шанса пьянства на период с 100 до 200 эпохи у всех водителей становился равным 25% (fig. 4). Во втором тесте со 150 эпохи стоимость ДТП увеличивалась в 100 раз (fig. 5).

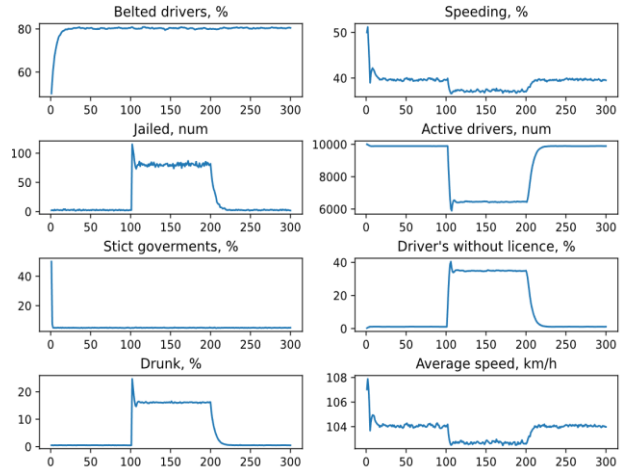


Fig. 4. Graphs showing key indicators during the stress test, when the probability of drunk driving for all drivers was 25% between epochs 100 and 200.

IV. Discussion and Conclusion

Using the current model as an example, we can observe the absence of a clear effect from increasing the period for depriving drivers of their licenses under Article 12.8. As can be seen, the system was clearly affected only by the change in the upper limit, and even then only in terms of the number of people deprived of their licenses, provided that the average number of drunk drivers in the period did not change. This finding confirms earlier studies that the fine policy does not affect the change in the frequency of incidents [4].

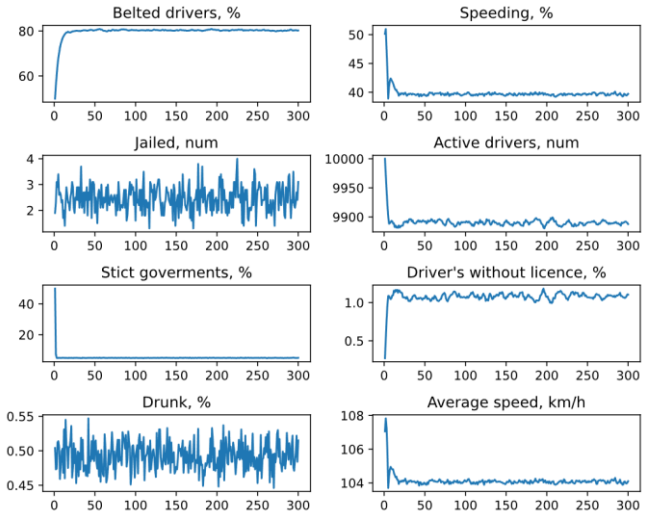


Fig. 5. Graphs for key indicators during the stress test, when after the 150th epoch the cost of a traffic accident increased in 100 times.

As part of the first stress test, where the likelihood of drunk driving for all drivers was assessed between the 100th and 200th eras, several interesting results emerged. First, it is worth noting that, as the likelihood of drunk driving increased, the rates of speeding and driving without a seatbelt did not change on average (moreover, the rate of speeding decreased). Second, the increase in the number of drunk drivers did not

lead to an increase in the number of strict government policies. Thirdly, as can be seen from the graph (fig. 4), after the 200th epoch, the population quickly returned to the dynamics that existed in the first 100 epochs. The second stress test showed that an increase in the cost of traffic accidents does not lead to a clear change in the behavior of the system's key indicators.

In the future, it is planned to reduce the number of such agents in order to create a mechanism for regional state policies that will be applied to a wide range of type D agents, which will make it possible to test various mechanisms for organizing penalty regulation: federal and unitary types. It is also planned to expand the current functionality of the model by adding a separate class of offenses: driving without a license. This will make it possible to track the dynamics of how often drivers get behind the wheel after having their license revoked, which factors will increase this likelihood, and which will decrease it. It is also expected to introduce a mechanism that will allow tracking the dynamics of recidivism among drivers who have previously had their licenses revoked or who have been sentenced to serve time in prison – the current model does not allow this.

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